

# Analog, safety and security

Slide 1 · 0–2 min

## Analog, safety and security

Every technique must discharge a named verification claim.

Bound the analog claim → Classify safety evidence → State the adversary

## Teaching script

Open with a decision the audience must make in this area. Ask which result they currently treat as sufficient evidence and keep that answer for the closing discussion.

The lesson selects a practical argument from the book. The study handout preserves the detailed examples, citations and worked artifacts.

Source: ch21:21–40, ch22:17–44, ch23:17–44, ch22:223–335

# Learning outcomes

Slide 2 · 2–5 min

## Learning outcomes

Demonstrate these decisions on a real plan row.

- Define an analog model's validity envelope.
- Distinguish unobserved from structurally irrelevant faults.
- State a threat model and its observation boundary.
- Preserve domain-specific qualifiers at sign-off.

## Teaching script

Read the outcomes as observable tasks. Ask learners to identify which one they can already demonstrate and which one currently depends on undocumented judgment.

At closure, require a concrete artifact or explanation for an outcome rather than a general expression of confidence.

Source: ch21:21–40, ch22:17–44, ch23:17–44, ch22:223–335

# Route through the lesson

Slide 3 · 5–7 min

## Route through the lesson

Four blocks, each ending with a decision.

- Bound the analog claim
- Classify safety evidence
- State the adversary
- Maintain specialized arguments

## Teaching script

Allow eighteen minutes for each block: a concept, its engineering consequence, a worked interpretation and a short discussion. Keep the discussion tied to the evidence available in the example.

Reserve the final eleven minutes for the assigned exercise and application to the learners' work. The timing is a course design allocation, not an engineering estimate.

Source: ch21:21–40, ch22:17–44, ch23:17–44, ch22:223–335

# An analog oracle needs an envelope

Slide 4 · 7–11 min

## An analog oracle needs an envelope

Acceptance depends on tolerance, time and operating conditions.

Operating conditions → Band and time window → Acceptance decision

### Teaching script

Replace a single exact expected value with a stated acceptable band over a stated observation window. Ask which operating conditions make that band meaningful.

The checker must distinguish an allowed transient from a settled violation. An unstated observation window can make two correct engineers reach different verdicts on the same waveform.

Source: ch21:21–40

# A fast real-number model omits physics

Slide 5 · 11–16 min

## A fast real-number model omits physics

Its value comes from a declared use boundary.

- Use RNM for represented digital–analog interactions.
- Do not infer unmodeled loading or continuous effects.

### Teaching script

List the behaviors represented by the real-number model and the physical effects it deliberately leaves out. Faster simulation is useful only for questions within that boundary.

A model that captures algorithm-level decisions can still omit current, loading and continuous dynamics relevant to another plan row. Route that row to appropriate evidence.

Source: ch21:41–76

# Correlation is a maintained obligation

Slide 6 · 16-21 min

## Correlation is a maintained obligation

A model comparison needs criteria set before observing the result.

- Compare at named ports and conditions.
- Record tolerances, validity and recheck triggers.

## Teaching script

Define the correlation campaign before looking at differences. Otherwise the tolerance can drift until the current model happens to pass.

Keep the model's validity record attached to the plan. A change in the implementation, operating range or intended use can require renewed correlation rather than inherited confidence.

Source: ch21:77-129, ch21:130-190, ch21:191-312

# Discussion: the model passed training

Slide 7 · 21–25 min

## Discussion: the model passed training

Name the evidence that would settle the decision.

- An RNM campaign validates the training algorithm.
- The next claim concerns physical channel variation.
- Can the original verdict discharge it?

## Teaching script

Expected answer: only if that variation is represented and validated within the model's intended-use envelope. Otherwise the plan needs separate physical evidence.

Ask learners to identify the missing mechanism rather than naming a slower tool by reflex. That mechanism determines the required model and observation.

Source: ch21:41–76, ch21:130–190

# Safety adds fault-response obligations

Slide 8 · 25–29 min

## Safety adds fault-response obligations

Correct intended behavior is only part of the argument.

- Separate systematic design defects from random hardware faults.
- Connect fault effects to detection and safe response.

## Teaching script

Explain the difference between implementing the wrong intended behavior and a physical fault perturbing a correctly designed mechanism. The verification activities overlap but the evidence questions differ.

Keep the lesson at the methodological level supported by the chapter. Its secondary-source discussion is not a substitute for reading the applicable safety standard for a product claim.

Source: ch22:17–44, ch22:45–84

# Not observed is not a safety classification

Slide 9 · 29–34 min

## Not observed is not a safety classification

A workload can miss a fault effect.

- Distinguish structural irrelevance from lack of excitation.
- Retain the argument behind any exclusion.

### Teaching script

Compare a fault outside all relevant observation cones with a fault that the chosen workload never activates. The first may have a structural justification; the second is an incomplete observation.

A database label cannot encode that distinction by itself. Require the evidence and its quantifier: all relevant behavior, or only the campaign's sampled workloads.

Source: ch22:85–160

# Detection has a deadline and a healthy case

Slide 10 · 34–39 min

## Detection has a deadline and a healthy case

A mechanism must react in time without inventing faults.

- Check fault detection and response timing.
- Check acceptable behavior without a fault.

## Teaching script

Follow the injected fault through detection, reporting and the required system reaction. Detecting an error too late may not satisfy the intended safety obligation.

Also run the healthy case. A mechanism that constantly signals an error is not validated by its ability to signal one during injection.

Source: ch22:161–222

# Discussion: unexcited means safe?

Slide 11 · 39–43 min

## Discussion: unexcited means safe?

Name the evidence that would settle the decision.

- A campaign never activates a modeled fault.
- The row is proposed for removal as harmless.
- What stronger argument would be needed?

## Teaching script

Expected answer: a justified structural or formal argument over the relevant behavior, rather than absence of an observed effect in the selected workload.

If that argument is unavailable, keep the fault and choose an explicit disposition: improve evidence, justify its handling, or address the exposed design gap as the chapter describes.

Source: ch22:85–160

# Security begins with a threat model

Slide 12 · 43–47 min

## Security begins with a threat model

Assets, attacker capability and trust boundaries define the claim.

Assets → Attacker capabilities → Trust boundaries

### Teaching script

Ask what the adversary can control and observe. A security property without that boundary may be either impossible or irrelevant to the product threat.

Distinguish an accidental functional defect from intentionally chosen inputs or sequences. Reusing a functional regression is useful evidence only for the attacks it actually models.

Source: ch23:17–44, ch23:45–117

# Functional correctness can coexist with leakage

Slide 13 · 47–52 min

## Functional correctness can coexist with leakage

A correct output does not settle every observation.

- Consider information flow and explicit exceptions.
- Treat release conditions as part of the policy.

### Teaching script

Use a design that computes the intended answer while exposing secret-dependent information elsewhere. A functional scoreboard can correctly pass that design.

An information-flow policy needs sources, sinks and permitted release conditions. Broad exceptions can remove the intended protection, so review their meaning instead of merely their syntax.

Source: ch23:118–170, ch23:171–209

# A model cannot expose a mechanism it omits

Slide 14 · 52-57 min

## A model cannot expose a mechanism it omits

Digital noninterference does not settle physical side channels.

- Choose observations that represent the attack.
- Allocate physical evidence for physical leakage.

### Teaching script

Explain why a digital model with no representation of power consumption cannot establish absence of power leakage. The issue is model scope, not whether the tool returned a proof.

A security campaign allocates evidence across functional, formal and physical methods. Keep the unmodeled observation explicit at the handoff.

Source: ch23:210-246, ch23:247-341

# Discussion: no path, no leak?

Slide 15 · 57–61 min

## Discussion: no path, no leak?

Name the evidence that would settle the decision.

- A structural report shows no direct secret-to-output connection.
- The threat includes timing observations.
- Which property remains unestablished?

## Teaching script

Expected answer: absence of secret-dependent observable timing behavior under the threat model. Structural disconnection is not the same as a noninterference argument.

Ask which model represents the timing observation and whether the policy permits any relevant release. The answer determines the appropriate check.

Source: ch23:171–209, ch23:210–246

# Qualification concerns a defined use

Slide 16 · 61–65 min

## Qualification concerns a defined use

Tool identity alone does not describe reliance.

- Record version, configuration and intended use.
- Identify the independent checks supporting reliance.

## Teaching script

Explain that confidence is attached to how a tool contributes to a specific process. A product name or generic qualification statement does not establish that this configuration and use are covered.

Keep the chapter's methodological treatment distinct from making a compliance determination. For an actual product, the applicable normative requirements need their own authoritative review.

Source: ch22:223–335

# A shared source can share an omission

Slide 17 · 65–70 min

## A shared source can share an omission

Independent evidence must challenge the original interpretation.

- Review model omissions against the claim.
- Revisit assumptions when intended use changes.

### Teaching script

Compare analog model validity, fault-effect analysis and a threat model. Each can be internally consistent while excluding the behavior that matters to the product.

The transferable practice is to expose the boundary and seek evidence that challenges it. It is not to merge these domains into one percentage or one universal checklist.

Source: ch21:130–190, ch22:45–84, ch23:45–117

# Close a specialized row with its qualifiers

Slide 18 · 70–75 min

## Close a specialized row with its qualifiers

Preserve the conditions under which the result is valid.

- Name the model, conditions and observations.
- Keep exclusions and residual claims visible.

### Teaching script

Have learners write a closure sentence for one selected domain. Require enough detail that a reviewer can say when the result would stop applying.

A change in operating envelope, fault model or attacker capability can reopen the row even when the implementation is unchanged. Record that re-evaluation trigger.

Source: ch21:130–190, ch22:85–160, ch23:247–341

# Discussion: one green status for every domain

Slide 19 · 75–79 min

## Discussion: one green status for every domain

Name the evidence that would settle the decision.

- Analog, safety and security reports are all green.
- Their excluded behaviors are absent from the dashboard.
- What must a sign-off reviewer recover?

## Teaching script

Expected answer: the model boundaries, observation conditions, exclusions and residual obligations behind each result.

Ask for one concrete counterexample to transfer: a physical effect absent from RNM, an unexcited fault or an attack observation omitted by the security model.

Source: ch21:130–190, ch22:85–160, ch23:247–341

# Exercise: two unobserved faults

Slide 20 · 79–85 min

## Exercise: two unobserved faults

Separate universal irrelevance from limited observation.

- One fault is outside every observation cone.
- Another is never excited by the workload.
- Decide which may be removed and how to handle the other.

## Teaching script

Exercise 22.4 (judgement). Two faults in a campaign's database carry the same row, marked not observed. One of them lies outside the cone of influence of every observation point. The other was never excited by the workload the campaign ran. Decide which of the two may be removed from the fault list, name the property of the evidence that separates them in §22.3's own terms, and name the three dispositions §22.3 leaves open for the fault that stays.

Assign book Exercise 22.4 in full. Require the exact property of the evidence and the three remaining dispositions from the chapter.

Expected answer: only the fault justified as unable to affect the relevant observations under any stimulus has the structural or formal basis for removal. The unexcited fault requires better stimulus, proof, or a recorded engineering judgment with a named owner.

Source: ch22:85–160

# What to change on Monday

Slide 21 · 85–87 min

## What to change on Monday

Make one existing verification decision reviewable.

- Add a validity trigger to a model record.
- Audit an unobserved-fault exclusion.
- Name a security observation missing from the current model.

## Teaching script

Ask each learner to choose one of these actions in an existing project. Require a small, inspectable deliverable and identify the person who will review it.

Keep the change connected to the lesson's evidence limits. A new dashboard, script or model is useful only if it strengthens the claim the plan row is meant to establish.

Source: ch21:21–40, ch22:17–44, ch23:17–44, ch22:223–335

# Scope and next reading

Slide 22 · 87–90 min

## Scope and next reading

Use the handout for the detailed artifacts and sources.

- Read the Analog, safety and security handout and book Chapters 21–23.
- Normative compliance and full analog or security tool flows are outside this lesson.

## Teaching script

Revisit the outcomes and ask the room to state a limitation of the strongest result discussed today. A complete answer identifies a model boundary, an assumption, a missing oracle or the scope of the underlying evidence.

Assign the detailed reading to the relevant chapters. The omitted topics are explicit scope boundaries, not claims that those topics are unimportant.

Source: ch21:21–40, ch22:17–44, ch23:17–44, ch22:223–335